



Module 4: Circuit Components

Semiconductors · Diodes · Digital ICs · Inductors · RF Packaging
2024-2028 Amateur Extra Question Pool



Module 4 Overview

- 6 exam questions — E6A through E6F — 12% of your Extra exam
- This module is heavier on identification than calculation
- Hour 1:
 - → E6A: Semiconductor materials and transistors
 - → E6B: Diode types and their applications
- Hour 2:
 - → E6C: Digital ICs — CMOS, logic gates, FPGAs
 - → E6D: Inductors, transformers, core materials, crystals
 - → E6E-F: RF semiconductors, IC packaging, electro-optics
- Focus areas: diode type identification is heavily tested. Know Zener vs. Schottky vs. PIN vs. varactor.



Part 1: Semiconductors and Transistors

Subelement E6A — 1 exam question



Semiconductor Materials

Materials

- Silicon (Si): most common; moderate electron mobility
- GaAs: higher electron mobility → microwave use
- GaN: highest electron mobility + power density → UHF/SHF power amps, highest MMIC freq
- N-type: excess free electrons (donor impurity)
- P-type: excess holes (acceptor impurity)
- PN junction reverse bias: depletion widens; no current

Transistor Types

- BJT: current-controlled; $\beta = \Delta I_c / \Delta I_b$
- NPN turns on at $V_{be} \sim 0.6-0.7V$
- FET: voltage-controlled; higher input impedance than BJT
- Depletion-mode FET: conducts with no gate voltage
- Enhancement-mode MOSFET: OFF with no gate voltage
- Dual-gate MOSFET: two gates for AGC or mixing
- MOSFET gate protection: Zener diodes clamp against ESD



BJT vs FET

- BJT (Bipolar Junction Transistor):
 - → Base current controls collector current
 - → Beta (h_{FE}) = collector current / base current
 - → Consumes input current — lower input impedance
 - → Good for high-frequency amplifiers; saturates easily
- FET (Field Effect Transistor):
 - → Gate VOLTAGE controls drain current
 - → Almost zero gate current — very high input impedance
 - → JFET: depletion type (reverse bias gate reduces current)
 - → MOSFET: enhancement (common) or depletion type
 - → Gate isolation: SiO₂ layer or PN junction



Part 2: Diode Types

Subelement E6B — 1 exam question



Diode Types and Applications

- Zener: conducts in reverse breakdown at a specific voltage — voltage regulation, clamping
- Schottky: metal-semiconductor junction
 - → Lower forward voltage drop ($\sim 0.2\text{-}0.4\text{V}$ vs $0.6\text{-}0.7\text{V}$ for silicon)
 - → Fast switching; VHF/UHF mixer and detector
- Varactor (varicap): PN junction used in reverse bias
 - → Junction capacitance varies with voltage — voltage-controlled capacitor
 - → Used in VCOs, AFC, phase modulators
- PIN diode: P-Intrinsic-N structure
 - → Low junction capacitance at RF
 - → Controlled by forward DC bias current — acts as variable RF resistor
 - → Used as RF switch or attenuator
- LED: forward voltage set by semiconductor band gap
- Point-contact diode: metal point on semiconductor — RF detector



Part 3: Digital ICs

Subelement E6C — 1 exam question



Logic Families and Gate Behavior

CMOS vs Others

- CMOS: lowest power consumption (near-zero static current)
- CMOS threshold: ~half supply voltage
- CMOS noise immunity: high (wide margin between 0 and 1)
- BiCMOS: CMOS input impedance + bipolar output drive
- Tri-state: three states — 0, 1, high-impedance (disconnected)
- Pull-up resistor: holds pin HIGH when open
- Pull-down resistor: holds pin LOW when open
- Comparator hysteresis: prevents chatter on noisy inputs

Gate Truth Tables

- NAND: output=0 only when ALL inputs are 1
- NOR: output=1 only when ALL inputs are 0
- OR: output=1 if ANY input is 1
- AND: output=1 only when ALL inputs are 1
- XOR: output=1 when inputs DIFFER
- XNOR: output=0 when inputs differ
- NOT: inverts the input
- FPGA: programmed with HDL (VHDL or Verilog)



Part 4: Inductors, Transformers, Crystals

Subelement E6D — 1 exam question



Core Materials and Transformer Basics

- Ferrite: high permeability → fewer turns needed for same inductance
 - → More temperature-sensitive than powdered iron
 - → Used for HF transformers, chokes, baluns
- Powdered iron: lower permeability; better temperature stability
 - → Used for HF inductors where frequency stability matters more
- Brass core: DECREASES inductance (diamagnetic material)
- Air core: no core loss, no saturation — VHF/UHF inductors
- Laminated cores: thin insulated layers reduce eddy current losses (power transformers)
- Toroidal core: confines magnetic field; low external radiation
- Magnetizing current: flows in primary even with no secondary load
- Inductor saturation: core flux exceeds capacity — inductance collapses
- Ferrite beads: suppress VHF/UHF parasitic oscillations on leads



Quartz Crystals and Piezoelectric Devices

- Quartz crystal equivalent circuit:
 - → Series RLC (the mechanical resonance) in PARALLEL with a shunt capacitor
 - → The shunt C is the electrode capacitance of the package
 - → Series resonance: lower frequency, lower impedance
 - → Parallel resonance: slightly higher frequency, very high impedance
- Piezoelectric effect:
 - → Mechanical stress → voltage
 - → Applied voltage → mechanical deformation
 - → Used in: quartz crystals, ceramic resonators, crystal filters, microphones, buzzers
- Permeability: determines how well a core material supports magnetic flux
 - → Higher permeability = more inductance per turn



Part 5: RF Semiconductors and Packaging

Subelements E6E-F — 1 exam question



RF and Microwave Semiconductors

- GaAs FET: high electron mobility; low noise figure — microwave LNA and oscillators
- GaN HEMT: highest electron mobility + power density
 - → Used for UHF/SHF power amplifiers
 - → GaN supports the HIGHEST frequencies in MMIC applications
- MMIC (Monolithic Microwave Integrated Circuit):
 - → Input and output impedance: 50 ohms
 - → Controlled gain and low noise figure
 - → DC power supplied through resistor or RF choke on the OUTPUT lead
 - → Never feed DC through the input — sensitive to any impedance change
- Microstrip: printed circuit board transmission line
 - → Impedance determined by trace width and substrate thickness
 - → Used to connect MMIC components with controlled impedance



IC Packaging and Electro-Optics

IC Packaging at RF

- SMD (Surface Mount): shortest leads, least parasitic inductance — best above HF
- DIP (Dual In-Line): through-hole; long leads = parasitic inductance; limited to HF
- SOP/SOT/QFP: surface mount variants
- Rule: smaller package = better at higher frequencies
- DIP is unusable at UHF due to lead inductance

Electro-Optical Devices

- Photovoltaic cell: light → electrical energy (photovoltaic effect); Si cell $\sim 0.5V$ open circuit
- Photoconductive material: more light = less resistance
- Optoisolator (optocoupler):
 - → LED + phototransistor in one package
 - → Complete galvanic isolation between circuits
 - → Used to interface logic to power circuits safely

Key Facts — Module 4

GaAs advantage	Higher electron mobility than Si
GaN application	Highest MMIC frequencies; UHF power amps
FET vs BJT	FET = higher input impedance
NPN turn-on voltage	$V_{be} \sim 0.6\text{--}0.7\text{V}$
Depletion-mode FET	Conducts with zero gate voltage
Zener diode	Constant voltage in reverse breakdown
Schottky diode	Low V_f ; metal-semiconductor; fast; VHF mixer
Varactor	Voltage-controlled capacitor (reverse biased)
PIN diode	RF switch; DC forward bias controls RF resistance
CMOS power	Near-zero static power
CMOS threshold	$\sim$ Half supply voltage
FPGA programmed with	HDL (Hardware Description Language)
Ferrite vs powdered iron	Ferrite = fewer turns; pow. iron = better temp stability
Brass core effect	DECREASES inductance

Crystal eq circuit

Series RLC || shunt capacitor

MMIC I/O impedances

50 ohms, DC fed through OUTPUT lead



Checkpoint Quiz 1 / 10

Which semiconductor material is used in microwave MMICs due to its higher electron mobility?

- A. Silicon
- B. Germanium
- C. GaAs
- D. Silicon carbide

Answer: C — GaAs has significantly higher electron mobility than silicon



Checkpoint Quiz 2 / 10

What is the primary advantage of a Schottky diode vs a silicon PN diode?

- A. Higher reverse breakdown voltage
- B. Lower forward voltage drop
- C. Higher junction capacitance
- D. Better temperature stability

Answer: B — Schottky: ~0.2-0.4V forward drop vs 0.6-0.7V for silicon PN



Checkpoint Quiz 3 / 10

What type of diode is used as a voltage-controlled capacitor?

- A. PIN diode
- B. Schottky
- C. Zener
- D. Varactor

Answer: D — Varactor: reverse-biased junction capacitance varies with voltage



Checkpoint Quiz 4 / 10

A PIN diode in an RF switch circuit is controlled by:

- A. RF signal amplitude
- B. Reverse bias voltage only
- C. Forward DC bias current
- D. Gate voltage

Answer: C — PIN diode: RF resistance controlled by forward DC bias current



Checkpoint Quiz 5 / 10

What logic family has the lowest static power consumption?

- A. TTL
- B. ECL
- C. CMOS
- D. BiCMOS

Answer: C — CMOS draws near-zero static current (only during switching transitions)



Checkpoint Quiz 6 / 10

Comparator hysteresis prevents what problem?

- A. Output saturation
- B. Oscillation of the oscillator stage
- C. Chatter from noise near the switching threshold
- D. Thermal runaway

Answer: C — Hysteresis creates two switching thresholds; prevents rapid toggling on noisy inputs



Checkpoint Quiz 7 / 10

A brass slug inserted into an inductor coil will:

- A. Increase inductance
- B. Decrease inductance
- C. Have no effect
- D. Increase Q

Answer: B — Brass is diamagnetic; it slightly opposes the magnetic field and decreases inductance



Checkpoint Quiz 8 / 10

What is the equivalent circuit of a quartz crystal?

- A. Parallel LC tank
- B. Series RLC only
- C. Series RLC in parallel with a capacitor
- D. RC low-pass filter

Answer: C — Series RLC (mechanical resonance) in parallel with shunt capacitor (electrode capacitance)



Checkpoint Quiz 9 / 10

How is DC power supplied to an MMIC?

- A. Through a coupling capacitor on the input
- B. Through a separate DC pin
- C. Through a resistor or RF choke on the output lead
- D. Through the ground plane

Answer: C — DC supplied through output lead via RF choke or resistor; input is left unaffected



Checkpoint Quiz 10 / 10

What does an optoisolator provide between its input and output circuits?

- A. Impedance matching
- B. Frequency translation
- C. Complete galvanic (electrical) isolation
- D. Voltage regulation

Answer: C — LED on one side, phototransistor on the other; no electrical connection between them



Before Module 5

Your assignment:

- Drill HamStudy E6A-E6F to 80%+ confidence
- Spend extra time on E6B (diode types) — Zener, Schottky, PIN, varactor are all tested
- Core materials (E6D): ferrite vs powdered iron, brass core — these are reliable exam questions

Coming up:

Module 5: Practical Circuits — the hardest module. Amplifier classes, filter responses, SDR/DSP, op-amp gain formulas, oscillator types. 8 exam questions. Budget maximum HamStudy time.